Structural Causal Models with Neural Mechanisms as an Alternative to Post-Hoc XAI Methods

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Abstract

Post-hoc explainability methods (e.g., SHAP, LIME) describe correlations in trained black-box models but cannot provide interventional analysis. We propose an explainable-by-design alternative: a structural causal model (SCM) whose mechanisms are learned with feed-forward neural networks (FFNNs) from data. An explicit directed acyclic graph (DAG) encodes domain assumptions each node can be estimated from its parents, enabling Pearl's *do*-operator for "what-if" analysis. We benchmark against a tuned XGBoost classifier using ROC AUC, PR-AUC, and F1. While XGBoost attains higher predictive performance, the SCM delivers causally grounded, auditable explanations and supports interventional analysis with transparent propagation through intermediate variables. We discuss limitations (DAG misspecification risk, chained error, limited tuning) and outline extensions (monotonic constraints, abduction-based counterfactuals, calibration, uncertainty). The results suggest SCM+FFNN is a possible alternative when explicit modeling of causal assumptions and interventional analysis are primary objectives.

GitHub — https://github.com/MSCA-DN-Digital-Finance/WP3_eXplainable_and_Fair_AI/tree/main/IRP1/Code/xai_project

Dataset — https://doi.org/10.5281/zenodo.11295916

Introduction

Current explainable AI (XAI) techniques in finance—such as SHAP or LIME—attempt to interpret opaque supervised machine learning (SML) models *after* training. These post-hoc explanations are limited to answering questions of *feature importance*, corresponding to the lowest level of Judea Pearl's *Ladder of Causality* (Pearl 2010).

According to Pearl, causality can be organized into three hierarchical levels:

1. **Association** (*What predicts what?*)
2. **Intervention** (*What if we do X?*)
3. **Counterfactuals** (*What would have happened if X had been different?*)

Standard SML+XAI methods remain confined to the first rung of Pearl's ladder: they describe *associations* without causal semantics. Consequently, they cannot reason about *interventions*, generate *counterfactuals*, or explain why a prediction changes in a way that is faithful to the data-generating process—limitations that matter in high-stakes domains such as loan approval.

Let $f : \mathbb{R}^d \to [0,1]$ be a supervised predictor trained to approximate the *observational* conditional $\hat{p}(y{=}1 \mid x)$ for binary $Y \in \{0,1\}$ and feature vector $x \in \mathbb{R}^d$. Post-hoc attribution methods (e.g., SHAP, LIME, Integrated Gradients) explain *values of f at observed inputs* by constructing an attribution vector $\phi(x) = (\phi_1(x), \ldots, \phi_d(x))$ such that

$$f(x) \;\approx\; \phi_0 \;+\; \sum_{j=1}^{d} \phi_j(x), \tag{1}$$

where ϕ_0 is a reference (baseline) and $\phi_j(x)$ is the contribution of feature x_j for this instance. These are summaries of *associations* learned by f.

Crucially, SML+XAI does not specify structural equations for how variables generate one another. Interventional queries require an SCM with mechanisms $X_i = f_i(\mathrm{Pa}_i, U_i)$ and the *do*-operator, which *replaces* the mechanism for intervened variables and propagates changes to descendants:

$$P\big(Y \mid \mathrm{do}(X_k{=}x')\big) \neq P\big(Y \mid X_k{=}x'\big).$$

Because post-hoc attributions operate on the fixed mapping $f(x)$ and lack these mechanisms, they cannot identify $P(Y \mid \mathrm{do}(\cdot))$ or individual counterfactuals; they remain associational by design.

Our goal is to develop and evaluate an *explainable-by-design* alternative to post-hoc XAI for loan approval. We propose a structural causal model (SCM) based on (Pearl 2009) whose mechanisms are parameterized by feed-forward neural networks (FFNNs), enabling (i) causal explanations grounded in an explicit directed acyclic graph (DAG), (ii) interventional "what-if" analysis via the *do*-operator, and (iii) comparable predictive performance to SML approaches. We empirically compare this SCM+FFNN approach against an SML baseline (XGBoost) on Lending Club data, assessing both predictive performance and the ability to support interventional analysis. This study makes the following contributions:

1. **Explainable-by-design causal architecture:** We instantiate a domain-informed DAG over credit features and learn node-specific mechanisms with FFNNs, yielding a transparent model whose assumptions are explicit and auditable.

2. **Interventional analysis for loan approval:** We demonstrate portfolio and individual "what-if" analyses using Pearl's do-operator (e.g., capping loan amounts or increasing income) and trace effects through intermediate variables (e.g., DTI).
3. **Comparative evaluation:** We benchmark SCM+FFNN against XGBoost using ROC AUC, PR-AUC, and F1, reporting both factual predictions and from-roots propagation to quantify accuracy versus causal capability.

Collectively, these contributions argue that SCM+FFNN might be a preferrable approach to post-hoc XAI layered on black-box SML.

Methodology

We use the Lending Club consumer credit dataset (Ariza-Garzón et al. 2024) with the following engineered feature set: *revenue* (annual income), *emp_length* (years), *fico_n* (FICO-like score), *loan_amnt*, *dti_n* (debt-to-income), *business* (purpose flag), *home_equity* (own/mortgage vs. rent), and the binary target *Default*. All features are standardized using train-set statistics.[1]

For our SCM, let $G = (V, E)$ be a directed acyclic graph (DAG) over variables V in our dataset. For each node $X_i \in V$ with parents $Pa_i \subset V$, the SCM specifies a structural equation

$$X_i = f_i(Pa_i, U_i), \tag{2}$$

where U_i is an exogenous noise term, mutually independent across nodes. Roots have $Pa_i = \varnothing$ and are observed. We approximate the unknown mechanisms f_i by feed-forward neural networks (FFNNs) f_i^θ with parameters θ and learn them from data through a supervised machine learning approach. For continuous nodes we use regression output layer with mean-squared error as loss function. For the binary outcome node Default we use a logistic output layer with binary cross-entropy as loss function. Inference under *factual* settings propagates observed values forward, practically this only requires the final mechanism $f_{Default}^\theta$ while inference *from roots* uses f_i^θ to compute the value of downstream nodes starting from observed roots. Interventional queries use Pearl's do-operator by replacing the corresponding mechanism(s) output with constants $P(\cdot \mid do(X_k = x^\star))$ and propagate values forward through the DAG.

We compare the SCM+FFNN to a SML baseline by training an XGBoost classifier on the same data and train/test split with standard tuning (RandomizedSearchCV over trees/learning-rate/columns-subsample, CV=5). Primary metrics are ROC AUC, PR-AUC, and F1 at threshold 0.5; for SCM we report both *factual* predictions and *from roots* predictions (all endogenous nodes recomputed via the DAG). We additionally present intervention analyses to demonstrate capabilities unavailable to post-hoc XAI.

Results

Figure 1 shows the assumed data-generating structure used by the SCM. Nodes are observed features in the Lending Club dataset and arrows encode direct causal dependencies. Each child node is modeled by an FFNN that takes its parents as inputs and is trained with the appropriate loss (MSE for continuous, BCE for the binary outcome). This graph provides the basis for prediction and interventional analyses.

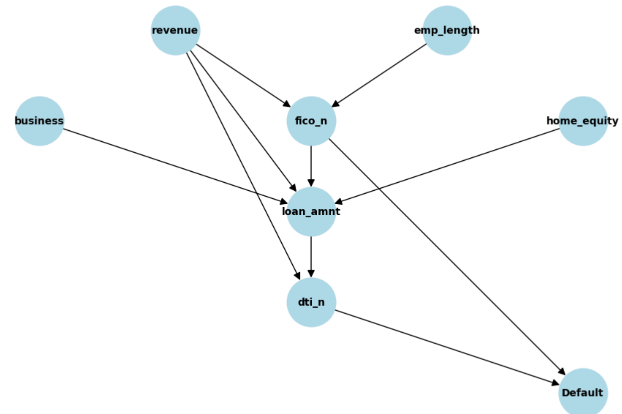

Figure 1: Directed acyclic graph (DAG) for the loan dataset. Nodes correspond to observed features (*revenue*, *emp_length*, *business*, *home_equity*, *fico_n*, *loan_amnt*, *dti_n*, *Default*). Arrows encode assumed causal dependencies between variables; each child node's mechanism (parents $\rightarrow$ child) is learned with a feed-forward neural network within the SCM.

Table 1 compares a tuned XGBoost classifier (SML baseline) with the proposed SCM+FFNN. On the test set, XGB attains the highest ROC AUC and PR-AUC (AUC $= 0.661$), while the SCM using *factual* intermediates is moderately lower (AUC $= 0.623$). When we recompute all endogenous variables *from roots* through the DAG, performance drops further (AUC $= 0.539$) likely due to error compounding along the causal chain and the absence of monotonic constraints. These results indicate that, under a pure predictive performance criterion, XGB is stronger. However, the SCM remains competitive while enabling causal analyses that SML+XAI cannot provide.

Model / Setting	AUC	PR-AUC	F1@0.5	Best F1	Best Thr.
XGB (5-fold CV, 50 cand.)	0.6614	0.3167	0.3902	0.3903	0.4994
SCM (factual)	0.6226	0.2730	0.3586	0.3609	0.4872
SCM (from roots)	0.5392	0.2129	0.3089	0.3290	0.4541

Table 1: Test performance comparison. "Factual" uses observed intermediates; "From roots" recomputes intermediates via the DAG before predicting *Default*.

SML+XAI's confinement to the associatonal level on the Ladder of Causality, also limits the interpretability of feature importance. Because *dti_n* is an engineered function of *loan_amnt* and *revenue*, these features are highly redundant in the tabular input space. XGBoost exploits all three, and SHAP assigns substantial importance to each of them (Fig-

[1]For further details on preprocessing and model training, we refer to the GitHub repository.

ure 2). From a purely statistical perspective this is consistent with how the model uses the features. However, from a causal and policy perspective it is problematic: the SHAP plot suggests three distinct levers for intervention, whereas in reality *dti_n* is a deterministic combination of income and loan amount. SHAP cannot disambiguate this redundancy or enforce that credit risk is mediated through DTI, while the SCM encodes this structure explicitly via the DAG.

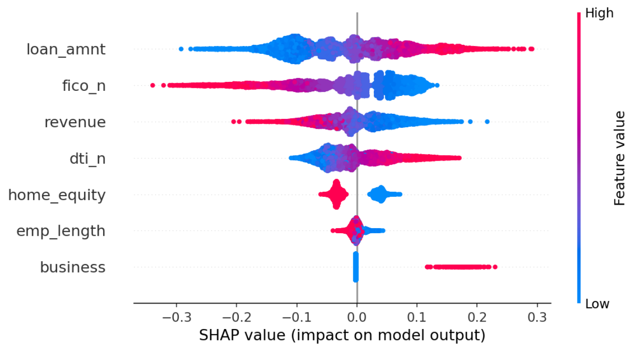

Figure 2: SHAP beeswarm plot for the XGBoost baseline on the loan dataset. Each point represents the SHAP value of a feature for one test instance, colored by the corresponding feature value. Features are ordered by mean absolute SHAP value, indicating their overall importance for predicting *Default*. The plot illustrates how the model distributes predictive influence across correlated predictors (*revenue*, *loan_amnt*, *dti_n*), reflecting associational feature usage rather than the mediated causal structure.

Table 2 illustrates interventions (Pearl's *do*-operator) for a test example (*scaled* feature space). Increasing income by $+0.2$ sd lowers the predicted default probability from 0.554 to 0.471 ($\Delta = -0.083$), consistent with its effect on the revenue $\rightarrow$ dti path. Increasing the loan amount by $+0.2$ sd raises risk to 0.603 ($\Delta = +0.049$) via higher predicted dti. Toggling *home_equity* from 1 to 0 increases risk to 0.594 ($\Delta = +0.040$); inspecting intermediate nodes confirms that, for this individual, the change shifts predicted dti upward despite a smaller predicted loan_amnt, highlighting the importance of mechanism constraints and local data support. These examples showcase how the SCM can answer policy-relevant questions of the form "What happens if we set X to x'?".

Intervention (Sample 10)	Factual (scaled)	New (scaled)	$P_{\text{fact}}(D)$	$P_{\text{do}}(D)$	Δ	Dir.
Income $+0.2$ sd	0.113	0.313	0.5540	0.4711	-0.0830	↓
Home equity $1 \rightarrow 0$	0.8605	-1.1621	0.5540	0.5942	$+0.0402$	↑
Loan amount $+0.2$ sd	0.562	0.762	0.5540	0.6032	$+0.0492$	↑

Table 2: Individualized interventions (do-operator) for Sample 10. Values in scaled space.

Conclusion

This work introduces an explainable-by-design loan approval approach that combines a domain-informed structural causal model (SCM) with feed-forward neural mechanisms (FFNNs). The key contributions are: (i) an explicit DAG over dataset variables with per-edge rationale, (ii) node-wise neural mechanisms learned directly from data, enabling end-to-end causal propagation, (iii) native support for policy-relevant *interventions* (*do*-operator), illustrated by concrete "what-if" analyses, and (iv) a transparent benchmark against a strong SML baseline (XGBoost) with standard metrics (AUC, PR-AUC, F1). Together, these artifacts map well to governance needs (traceable assumptions, stress-testing, and auditable logic) that post-hoc XAI cannot fully satisfy.

The current SCM shows a modest accuracy gap to XGB on test data. Likely causes include restricted parent sets at the outcome, error compounding when predicting from roots, limited hyperparameter tuning, and absence of explicit monotonicity constraints. The DAG may be misspecified in places, and the present implementation does not perform formal *counterfactual* abduction (residual preservation) for individual recourse. Calibration of individual FFNNs and statistical uncertainty of performance results were not reported.

Future work could address (a) abduction–action–prediction for counterfactual recourse with feasibility/cost constraints; (b) monotonic priors on mechanisms; (c) strengthen validation with cross-validated per-node training, HP sweeps, and uncertainty quantification for performance results; (d) policy-shift/OOD experiments and fairness/path-specific analyses; and (e) compliance mapping to EU AI Act and ECB/EBA guidance. These extensions could aim to close residual accuracy gaps while building on the SCM's core advantage: causally grounded, actionable, and auditable decision support for loan approval.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT in order to improve readability and language. After using this tool, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content.

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